Profile

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Education details

Degree/Certificate	Institution	Percentage/CGPA
B-Tech	IIT BHU, Varanasi	7.78
CBSE-XII	JNV, Unnao	87.6
CBSE-X	JNV, Unnao	91.20

Certifications/ Trainings

- NVIDIA Certified Associate – Generative AI & LLMs
- NVIDIA AI Enterprise for Azure Professionals
- AI for All: From Basics to GenAI Practice
- NVIDIA AI Technical Curriculum (2024, 2025)

Overall Experience

Languages - Python, C++.

Statistical Tools – Pandas, Numpy, Mapper (TDA), SciPy.
Data visualization Tools – Matplotlib, Seaborn, Plotly.
Imaging Tools – OpenCV , Pillow.
UI Tools – Gradio, Streamlit.
Hardware Platforms - Nvidia Jetson (AGX Orin, Xavier, Nano, TX2), Nvidia dGPU machine (H200, H100, V100, A100 and etc), Intel Gaudi3, Intel ASUS NUC 15 Pro+ Edge device, AMD EPYC CPUs.
Nvidia Tech Stack – Nemo, DeepStream, TensorRT, TAO, Riva, Kaolin and etc.
Intel Tech Stack – Optimum-Habana, OpenVINO, DL Streamer and etc.

Machine Learning Tools – Tensorflow, Pytorch, Scikit-learn, TensorRT.
MLOps Tools and Technologies – Mlflow, Airflow, DVC & CML, MLEM, Git, Github, Github actions, Feast, Docker, Vscode and etc.
AWS Tools and Technologies – EC2, ECS, S3, Sagemaker, Bedrock, ECR, Redshift & dynmoDB.
Generative AI Tools and Technologies – Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) and Advanced RAG techniques, Parameter-efficient fine-tuning (LoRA, QLoRA), Inference optimization and serving using vLLM, Llama-cpp, Ollama, TensorRT-LLM, and NVIDIA NIM, LLM quantization, DeepSpeed, Agentic, Multimodal LLMs and RAG, Diffusion Models, Prompt Engineering, LangChain, LlamaIndex, and LLMOps.

- 4.5 years of hands-on experience in **Artificial Intelligence**, spanning Computer Vision, MLOps, LLMOps, and Generative AI systems.
- Experience in **computer vision applications**, including image classification, object detection, multi-object tracking, segmentation, and **real-time video analytics at the edge**.
- Hands-on MLOps CI/CD pipelines on AWS Sagemaker.
- Hands-on Implementing and optimizing MLOps and LLMOps CI/CD/CT/CM pipelines.
- Hands-on experience with **ML pipeline orchestration and automation**, including feature engineering, experiment tracking, model versioning, and deployment workflows.
- Managed ML feature storing and serving using Feast and AWS.
- Hands-on Trustworthy AI (Fair AI, Explainable AI, Secured AI and Ethical AI).
- Hands-on experience building **Generative AI applications**, including **RAG, Multimodal RAG, agentic workflows, and LLM-powered systems**.
- Experience in **LLM fine-tuning and inference optimization** using **PEFT techniques (LoRA, QLoRA)**, **DeepSpeed**, **TensorRT-LLM**, and hardware-aware optimizations.
- Hands-on developing GenAI SDKs using Langchain, Llamaindex and Semantic-kernel frameworks.
- Experience on working with FAST and Flask APIs.

	• Experience on working with PostgreSQL database. • Hands-on AutoML. • Explored topological data analysis on image and tabular data. • Hands-on experience with **containerization and Docker-based deployment** of ML and AI services

PROJECTS		
Employer Name	**Roles & Responsibilities**	
LTM	**Project: Transit Watch – Edge AI Metro Safety & Video Analytics** **Key Responsibilities:** • Developed a real-time edge AI video analytics solution for metro safety and operational monitoring using live CCTV streams. • Built computer vision pipelines to detect events such as yellow line crossing, intrusion, unattended objects, wrong-direction movement and etc. • Designed and optimized edge deployments tailored to specific hardware architectures, including Intel® Core™ Ultra–based ASUS NUC Pro 15+ and NVIDIA Jetson AGX Orin device. • Leveraged Intel® OpenVINO™ to utilize CPU, GPU, and NPU acceleration on Intel platform, and NVIDIA TensorRT for GPU-accelerated, low-latency inference on Jetson. **Tools and Technology used:** **Tools –** Python, Intel® OpenVINO™, Intel DL Streamer, NVIDIA TensorRT, DeepStream SDK, FFmpeg, GStreamer, OpenCV, FastAPI, PostgreSQL and etc. **Technology** – Edge AI, Real-time Computer Vision, Video Analytics, Hardware-Aware Optimization, Low-latency Inference, Edge Deployment.	July2025-Current
LTM	**Project: PAN 2.0 – AI-Based Identity Deduplication & Verification (PoC)** **Key Responsibilities:** • Built API-driven deduplication pipeline to validate PAN applications using photo, signature, and textual attributes. • Implemented face and signature similarity validation using deep learning–based embeddings and cosine similarity matching. • Designed multi-modal verification workflows combining image and text signals with configurable similarity thresholds. • Generated similarity scores and insights for downstream decision and reporting systems. • Deployed containerized inference and validation services on AWS ECS. **Tools and Technology used:** **Tools –** Python, DeepFace, TensorFlow, PyTorch, Sentence Transformers, FAST APIs, Docker, AWS EC2, AWS ECS, Git, GitHub, Linux. **Technology –** Face Recognition, Signature Embedding & Matching, Metric Learning, Multi-modal Feature Extraction, Vector Similarity Search, Cosine Similarity Scoring, Threshold-based Decisioning, Cloud-native AI Deployment.	March2025 – May2025
LTM	**Project: LLM Factory – Hardware-Aware LLM Fine-Tuning, Optimization & Serving Platform**	Nov2024 – June2025

	Key Responsibilities: • Built an end-to-end platform for LLM fine-tuning, optimization, and serving across diverse hardware environments. • Integrated multiple LLM frameworks including NVIDIA NeMo, Hugging Face, Unsloth, Intel Optimum-Habana, and InstructLab. • Designed training pipelines with evaluation, logging, checkpointing, and resource monitoring. • Enabled flexible LLM serving using NVIDIA NIM, TensorRT, OpenVINO™, vLLM, llama.cpp, and Ollama. **Tools and Technology used:** **Tools –** Python, NVIDIA NeMo, Hugging Face, Unsloth, Intel Optimum-Habana, InstructLab, NVIDIA NIM, TensorRT, OpenVINO™, vLLM, llama.cpp, Ollama. **Technology –** LLM Fine-Tuning, Hardware-Aware Optimization, Quantization, PEFT, Model Evaluation, Inference Optimization.	
HCLTech	**Intel GenOps Project: Automated Data Crawler, Ingestion, and Retrieval Framework** **Project Overview**: Developed an automated system leveraging generative AI and Retrieval-Augmented Generation (RAG) to efficiently extract, ingest, and retrieve data from diverse sources such as Wiki pages, SharePoint, and various web URLs, including privileged Intel data. This framework utilized specialized crawlers, a Data Ingestion pipeline, and a Retriever pipeline to streamline data collection and access. **Key Responsibilities**: • **Custom Crawlers**: Designed and implemented tailored crawlers for each data source to ensure efficient data extraction while adhering to access protocols. • **Data Ingestion pipeline**: Developed a Data Ingestion pipeline to facilitate the transfer, organization, and storage of collected data, ensuring seamless integration with retrieval processes. • **Retriever pipeline**: Created a Retriever pipeline for vector-based searches, enabling quick and relevant information retrieval based on user queries. **Tools and Technology used:** Apache Airflow, Azure Cloud Platform, MinIO, GitHub, PostgreSQL, Langchain/Llamaindex SDK, Generative AI, Retrieval-Augmented Generation (RAG), and APIs.	June2024 – Oct2024
HCLTech	**Project: Dynamic MLOps best practices incorporating Responsible AI functionalities** **Key Responsibilities:** • Implementing and optimising MLOps (managing the lifecycle of ML/DL models, including development, deployment, and maintenance). • Ensuring deployed model's fairness, explainability, security and robustness. • Implementing and optimizing LLMOps (managing the lifecycle of LLM and LLM-powered applications, including development, deployment, and maintenance). **Tools and Technology used:** Tools – Python, Mlflow , DVC, CML, MLEM, Git, Github, Github actions, Tensorflow, Pytorch, AWS S3 and EC2 and Vscode. Technology – Data Ingestion, Data Processing, supervised Machine Learning, Model Monitoring, AutoML, Hyperparameter tuning, PEFT (lora and qlora), Prompt engineering, Deepspeed, AGI,	01Feb2023 – May2024

	multimodal LLM, RAG, multimodal RAG and TensorRT-LLM.	
HCLTech	**Project: Trustworthy/Responsible AI** **Key Responsibilities:** Created framework to assess AI models with respect to different principles of responsible AI along with quantification of each. It supports evaluation with respect to transparency/explainability, fairness, security & privacy, ethical, and robustness for Machine Learning/Deep Learning models. At the end it calculates the overall trustworthy/robust score which tells user how much trustworthy the assessed model is. **Tools and Technology used:** Tools – Tensorflow, Pytorch, Scikit-leran, aif360, dalex, adversarial-robustness-toolbox, fairlearn. Technology – Data Processing, Supervised Machine Learning, Neural networks, Data Visualization and etc.	01Aug2022 – Jan2023
HCLTech	**Project: Conduent camera enforcement software development** **Key Responsibilities:** • **Detected traffic offenses:** Applied AI techniques to identify seatbelt non-compliance, cellphone use while driving, illegal turns, box junction blocking, and tailgating. • **Developed a processing pipeline:** Built a pipeline to process traffic video feeds, crop vehicle windshield regions, and forward the cropped images to an object detection model. • **Implemented object detection and tracking:** Leveraged OpenCV-based object detection and tracking to accurately detect and monitor traffic offenses from video streams. • **Optimized solutions for NVIDIA Jetson:** Improved the performance of all solutions to ensure efficient execution on NVIDIA Jetson hardware. • **Conducted benchmarks:** Evaluated base models against TensorRT-optimized models on NVIDIA Jetson, demonstrating significant performance gains. **Tools and Technology used:** Devices – Nvidia Jetson AGX Orin and Xavier. Tools – Python, C++, Tensorflow, Darknet, Pytorch, TensorRT Technology – Object detection & tracking, Image classifcation, Predictive Modelling, Supervised Machine Learning, Neural Networks, Model optimization.	01Oct2021 – 30Dec2022
HCLTech	**Project: Topological data analysis on image data** **Key Responsibilities:** Practice TDA approaches for the image dataset with support of deep learning network to have deep insight of convolutional neural network through weights and feature maps visualization. We had formulated three basic flows to handle image datasets with the TDA and validated our proposed approaches by taking the datasets. **Tools and Technology used:** Tools – Python, Mapper, Tensorflow, Pytorch. Technology –Data Ingestion, Data Processing, Supervised Machine Learning, Neural networks, Data Visualization and etc.	01June2021 – 30Sep2021